

Level Crossings

Driver Revision Guide

Use of the Warning Horn at Level Crossings

Sources: GERT8000-TW8 Issue 10 (Level crossings – drivers' instructions); GERT8000-TW7 Issue 10 §4.2 (Wrong-direction movements: level crossings); GERT8000-TW1 Issue 22 §48 (Warning horn); GERT8000-TW5 Issue 14 §23 (Track circuit actuators); GERT8000-SP Issue 7 (Speeds – differential speed definition); RS521 Issue 9 §5.13 and §6 (Whistle boards; Level crossing signs and indicators); and the EMR Level Crossings & Use of Warning Horn training presentation.

1. Types of Level Crossing

GERT8000-TW8 §1 groups every level crossing you will meet into four families. The locations of controlled, automatic, open and traincrew-operated crossings are listed in Table A of the Sectional Appendix.

Automatic crossings

- AHBC – Automatic half-barrier crossing
- ABCL – Automatic barrier crossing, locally monitored
- AOCL – Automatic open crossing, locally monitored
- R/G – Crossing with red and green warning lights (the red and green lights are for road users; R/G crossings can also be user-worked – see below)

Controlled crossings

- At the location: MCB – manually-controlled crossing with barriers; MG – manually-controlled crossing with gates
- Remotely: RC – remotely controlled crossing with barriers; CCTV – barrier crossing with closed-circuit television; OD – barrier crossing with obstacle detection; TMO – traincrew-operated crossing

Open crossings

- OC – crossing without barriers, gates or road warning lights
- Barrow or foot crossing with white light indicators

User-worked crossings

- R/G – crossing with red and green warning lights (also classed as automatic – see above)
- UWC – Occupation and accommodation crossing (including bridleway crossings)

Some automatic crossings can also be operated by trains making wrong-direction movements. These are identified in the Sectional Appendix by the suffix letter X, for example AHBC-X or ABCL-X (TW8 §1; controls are provided for movements in both directions on single and bi-directional lines).



An AHBC at a country road – barriers, road signals and the driver's crossing indicator are all visible to the right of the signal box.



An AOCL/ABCL-type crossing showing the half-barriers down and the red St Andrew's cross road signals illuminated.



A manually-controlled crossing (MCB/MG type) protected by a signal worked from the adjacent signal box.



Obstacle detection equipment at an MCB: radar provides the primary rail-safety check; a laser (LIDAR) scan adds pedestrian protection. Signaller indications and manual over-rides are provided for degraded working when CCTV is unavailable.



A user-worked occupation/accommodation crossing (UWC) with user-operated gates.



A user-worked crossing with a telephone for contacting the signaller and a Stop sign for the crossing user.

2. Drivers' General Instructions (TW8 §2)

2.1 Reporting equipment failure

You must report to the signaller, in the quickest way possible, any defect or irregularity with level crossing equipment.

2.2 Carrying out the signaller's instructions

- If the signaller tells you to approach a crossing at caution, you must not pass over it until you have made sure it is safe to do so.
- If the signaller tells you to, you must report back whether the crossing is safe for the passage of trains.
- If the signaller asks, you must tell them whether or not: the barriers are fully lowered; the crossing is clear.

2.3 Vehicle gates left open

You must report to the signaller, in the quickest way possible, any level crossing gates which have been left open.

2.4 Passing over level crossings under local control

You must approach the crossing at caution and pass over it only if a green handsignal is shown at the crossing, when you have been told:

- an AHBC is under local control, or
- to make a wrong-direction movement over a CCTV, OD or RC crossing that is under local control, or
- to make a movement in either direction over a CCTV, OD or RC crossing that is under local control on a line under possession.

Driver experience – Level crossings under local control

If a crossing fails, the signal protecting it is held at danger and cannot be cleared. After contacting the signaller you will be instructed to pass the signal at danger and proceed towards the crossing. At the crossing, an attendant will give you a green handsignal to authorise you over the crossing once it is safe. After receiving the green handsignal you may pass over the crossing and continue at caution through the section to the next signal.

3. AHBC Crossings and Crossings Operated by a Crossing Keeper (TW8 §3)

If your train fails on the approach to one of these crossings and the signaller reminds you about the level crossing, you must assure the signaller that you will make no further movement until the signaller authorises it.

Background (training material): an AHBC is activated by the train striking a treadle in the track. At line speed it takes around 27 seconds from activation to the train reaching the crossing; warning lights and the alarm start, and the barriers close, before the train arrives. The barriers re-open once the rear of the train has cleared the crossing.

4. ABCL and AOCL Crossings (TW8 §4)

4.1 If a train is not required to stop at the crossing

- On passing the warning board, control the speed of your train to not more than the speed shown on the speed restriction board (or driver machine interface).
- Where differential speeds are shown, control your speed to comply with the speed that applies to your train between the board and the crossing (see Section 8).

- On passing the speed restriction board or sighting board, make sure you can see the crossing is clear and that the white light next to the crossing is flashing.
- You may then proceed to the crossing at no more than the speed shown, and accelerate as soon as the front of the train is on the crossing.



Sighting/speed-restriction boards on the approach to an ABCL – here showing a differential 45 mph restriction (see Section 8).

4.2 If a train is required to stop at the crossing

- On passing the warning board, control the speed of your train to stop at the stop board.
- After stopping: if there is a plunger, operate it to activate the road-traffic signals, but not before you are ready to restart.
- Make sure you can see the crossing is clear and the white light next to the crossing is flashing.
- Sound a short blast of the horn as a warning of the movement (only between 0600 and 2359, except in an emergency or when anyone is on or near the line).
- Restart your train and proceed over the crossing.

4.3 Train delayed or stopped out of course on the approach

- If delayed after the white light has started flashing, approach the crossing at caution even if the white light continues to flash.
- If the white light is still flashing when your train reaches the crossing, you may pass over without stopping.
- If the white light has stopped flashing when your train reaches the crossing, you must stop short of it and carry out the instructions in 4.4 below.

4.4 When a train must be stopped

You must stop before reaching the crossing if:

- the white light next to the crossing is not flashing, or the red light is flashing
- the crossing is obstructed
- you cannot see if the crossing is clear
- the signaller has told you the crossing has failed
- you have been told the road-traffic signals have been switched off and, at an ABCL, the barriers have been left raised.

You may then pass over the crossing as long as you make sure it is safe to do so, and sound the horn continuously until the front of your train is on the crossing.

Where an emergency plunger is provided

Use the emergency plunger to operate the crossing controls when the equipment has failed, or has stopped working because it has been operating for an unusually long time. This must be operated even if the barriers are already down and the road lights are flashing – it does not start the white light sequence. After operating the plunger you may pass over the crossing, but before doing so you must make sure it is safe to do so and sound the horn continuously until the front of your train is on the crossing. Report your actions to the signaller and Control at the first opportunity.

Passage during darkness

If the white light at the crossing is not flashing, you must not pass over the crossing during darkness unless one of the following applies: your train is a passenger or empty-coaching-stock train with the interior lights lit; arrangements have been made to prevent road traffic passing over the crossing; or, at an ABCL, the barriers are in the lowered position with the lights on the barriers lit.

Driver experience – failure to operate the plunger

A driver approaching an ABCL did not see the white flashing light and did not operate the plunger before passing over the crossing, passing the red crossing light. This kind of failure to follow the ABCL/AOCL stopping procedure is treated as a serious operational incident and can lead to the driver being investigated.

Driver experience – wrong-direction working over an AHBC-X

A train failure required a wrong-direction movement over an AHBC fitted with wrong-direction controls (AHBC-X). The reverse movement, made with permission, was inside the wrong-direction trigger point, and the crossing display in the signal box showed it had failed. The driver saw the crossing was not working correctly but judged it safe to cross anyway. The incident was recorded as a miscommunication between signaller and driver – a reminder that the driver must be satisfied a crossing is genuinely safe, not assume it from a signaller's general authority.

5. Open Crossings (TW8 §5)

5.1 If a train is not required to stop at the crossing

- On passing the warning board, control the speed of your train to comply with the speed shown on the combined speed and whistle board, maintained between the board and the crossing.
- Where differential speeds are shown, control your speed to comply with the speed that applies to your train (see Section 8).
- On passing the combined speed and whistle board, make sure you can see the crossing is clear.

- You may then proceed at no more than the speed shown, and accelerate as soon as the front of the train is on the crossing.
- You must stop before reaching the crossing if it is obstructed or you cannot see whether it is clear. You may then pass over only once you have made sure it is safe to do so, sounding the horn continuously until the front of your train is on the crossing.

5.2 If a train is required to stop at the crossing

On passing the warning board, control the speed of your train to stop at the stop board. Before passing the stop board, make sure it is safe to pass over the crossing.

6. Wrong-Direction Movements at Level Crossings (TW7 §4.2)

Wrong-direction speed restriction boards are positioned on the approach to crossings that have wrong-direction controls (identified in the Sectional Appendix by the suffix X). You must control your speed to comply with the figure shown, then accelerate to the appropriate speed once the front of the train is on the crossing.



A wrong-direction speed restriction board. The X identifies that the crossing ahead has wrong-direction controls; the number is the permissible speed (mph or km/h) up to the crossing.

AHBC

If the signaller tells you the crossing is being locally operated, approach at caution and do not pass over unless authorised by a green handsignal shown at the crossing.

ABCL and AOCL

Carry out the same approach-at-caution, plunger and horn rules as for a normal-direction movement; if an emergency plunger is provided you must use it before crossing, and you may pass over once you have made sure it is safe and sounded the horn continuously until the front of the train is on the crossing.

CCTV, OD or RC crossings

- No attendant at the crossing: stop opposite the signal protecting the crossing on the other line and get further instructions from the signaller (during single line working, the pilot will tell you to approach at caution and not pass over until you are sure it is safe).
- Attendant on duty: approach at caution and only pass over when authorised by a green handsignal.

Manually-controlled crossings

Approach at caution and do not pass over until you are sure it is safe to do so.

Crossing with red and green warning lights (R/G)

Unless the crossing has wrong-direction controls, you must: approach the crossing at caution; stop short of the crossing; sound a short blast of the horn as a warning of the movement; pass over only if it is safe to do so. If any crossing within the limits of the wrong-direction movement has wrong-direction controls, you do not have to carry out these instructions for that crossing once the movement has been completed or the train has passed a wrong-direction speed restriction board.

Barrow or foot crossing with white-light indications

If you are told the crossing will not operate normally for the movement, approach at caution and check it is safe before passing over.

7. Use of the Warning Horn (TW1 §48)

General rule: you must only use the horn as much as is necessary to give an effective warning or to make sure safe working takes place.

Sounds to use

Circumstances	Sound to use
To give a warning of your train approaching	Both tones if available
To give an urgent warning of immediate danger	Long blast
Train in distress	Continuous series of long blasts
When passing a whistle board	Low tone if available
To give a warning of a train movement	Short blast (soft setting or depot whistle if in a depot/siding, or entering a shed/building)
To sound a local or special code	Follow local or company instructions

Whistle boards

- You must only sound the horn when passing a whistle board between 0600 and 2359, except in an emergency or when anyone is on or near the line.
- Sound the horn on the low tone if available as you pass the whistle board, and not before.
- If making a wrong-direction movement, sound the horn when passing a whistle board that applies to movements in the right direction.



A whistle board (RS521 §5.13) – a retro-reflective sign or cut-out marking where the horn must be sounded on the approach to a level crossing.

Urgent warning and train in distress

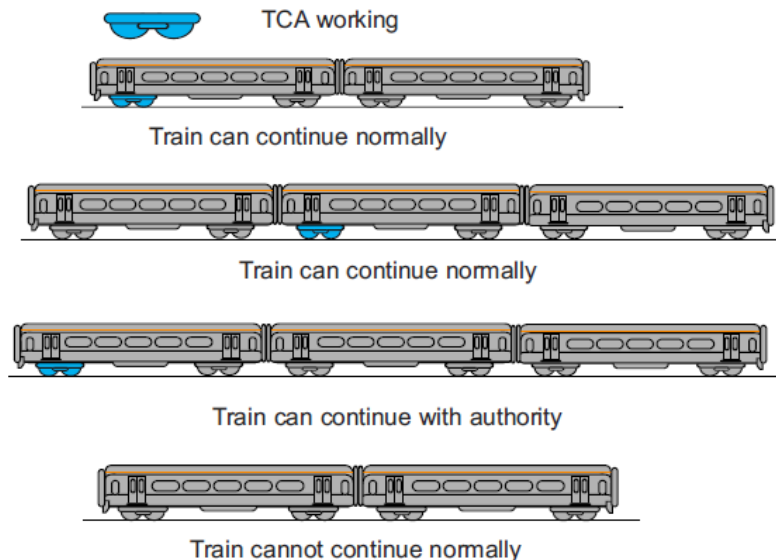
- To give an urgent warning of immediate danger: sound a long blast to warn the driver of another train in danger, or anyone on or dangerously near the line who does not acknowledge by raising one arm above the head, or does not appear to move clear.
- Train in distress: give a continuous series of long blasts.
- General: sound the horn at any other time you consider necessary, or when specified in your company instructions.

8. Differential Speeds at Level Crossings (SP §1)

Where a differential permissible speed is shown on a speed restriction board or combined speed and whistle board, the higher speed applies to: passenger trains (loaded or empty), parcels and postal trains (loaded or empty), and light locomotives. The lower speed applies to all other trains.

9. Track Circuit Actuators and Level Crossings (TW5 §23)

A track circuit actuator (TCA) is fitted to some vehicles to help operate track circuits, including those that detect a train's approach to some automatic level crossings.



TCA working: a train of one or two vehicles needs at least one working TCA on the train; a train of three or more vehicles needs at least one working TCA on either of the first two vehicles and at least one working TCA on either of the last two vehicles.

- Train can continue normally: at least one TCA working on a one/two-vehicle train, or one working on either of the first two and either of the last two vehicles on a longer train.
- Train can continue with authority: fewer TCAs working than above, but at least one TCA working on the train and authority given by the train operator's control.
- Train cannot continue normally: no TCA working that meets the requirements above and no authority obtainable.

If the train cannot continue normally

You must not move the train until instructed to do so. Except for an AHBC fitted with treadles, the signaller must instruct you to approach at caution, and not pass over until sure it is safe to do so, any automatic level crossing or barrow/foot crossing with white-light indications. If told to approach a level crossing at caution for this reason, sound the warning horn continuously from where you start to travel at caution until the front of the train is on the crossing. When given authority to proceed, you can do so at normal speed.

Quick-Reference Questions and Answers

Question	Answer
What must you do if your train fails on the approach to an AHBC and the signaller reminds you of the level crossing?	Assure the signaller that you will make no further movement until the signaller authorises it.
At an ABCL/AOCL, what two things must you check before proceeding past the speed restriction or sighting board?	That the crossing is clear, and that the white light next to the crossing is flashing.
Between what hours must you sound the horn when passing a whistle board (except in an emergency or when anyone is on or near the line)?	0600 and 2359.
What sound must you use when passing a whistle board, if available?	The low tone.
If the white light at an ABCL/AOCL has stopped flashing when your train reaches the crossing, what must you do?	Stop short of the crossing and carry out the "when a train must be stopped" instructions (make sure it is safe, then sound the horn continuously until the front of the train is on the crossing).
When must an emergency	Whenever the level crossing equipment has failed or has stopped

Question	Answer
plunger be operated, even if the barriers are already down and the road lights are flashing?	working because it has been operating for an unusually long time – the plunger must still be operated even though it will not start the white light sequence.
Under what three circumstances can a train pass over a crossing with the white light not flashing during darkness?	The train is a passenger or empty-coaching-stock train with interior lights lit; arrangements have been made to prevent road traffic passing over the crossing; or, at an ABCL, the barriers are lowered with the lights on the barriers lit.
What does the letter X mean as a suffix on a level crossing type in the Sectional Appendix (e.g. AHBC-X)?	It identifies that the crossing has wrong-direction controls provided.
When making a wrong-direction movement over a CCTV, OD or RC crossing with no attendant present, what must you do?	Stop opposite the signal protecting the crossing on the other line and get further instructions from the signaller.
When making a wrong-direction movement over a crossing with red and green warning lights (no wrong-direction controls), what four things must you do?	Approach at caution; stop short of the crossing; sound a short blast of the horn as a warning of the movement; pass over only if it is safe to do so.
On a differential speed restriction board at a level crossing, which trains get the higher figure?	Passenger trains (loaded or empty), parcels and postal trains (loaded or empty), and light locomotives. The lower figure applies to all other trains.
How many TCAs must be working on a train of three or more vehicles for it to continue normally?	At least one working on either of the first two vehicles, and at least one working on either of the last two vehicles.
If a TCA fault means the train cannot continue normally, how must you approach automatic level crossings and barrow/foot crossings with white lights (except an AHBC with treadles)?	At caution, sounding the horn continuously from the start of the caution approach until the front of the train is on the crossing, and not passing over until sure it is safe.
What must you report to the signaller about level crossing gates?	Any level crossing gates which have been left open, reported in the quickest way possible.
Under local control, when may you pass over an AHBC, or make	Only when a green handsignal is shown at the crossing, after approaching at caution.

Question	Answer
a wrong-direction movement over a CCTV/OD/RC crossing under local control?	